THE ACTION OF PSL_2(Z) ON THE ALGEBRA LAMBDA IS FAITHFUL

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Allowed standard prerequisites (editorial).

- The matrices p and s generating the action, with s of the form s_0 + s_1 lambda. Same source, Definition labelled def:ps; invoked inside the counted proof at lines 858-883. Lets a word in s and p^{+-1} be read as a polynomial in lambda with coefficients in 2x2 matrices.
- The determinant and trace formulas for the elements of Lambda. Same source, Lemma labelled lem:4, clause (i); invoked inside the counted proof at lines 900-903. Converts the centrality equation eta = kappa I into the identity (-lambda)^ell = kappa^2, which forces ell to be even.

Source reference. Original definition 1.14: sources/sigma-2011-069-askey-wilson/pinned-original-source.tex, source lines 858–858. Navigation pointer for \ref{def:ps} used inside the counted span (line 858); the argument that consumes it is retained in the span.

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Source reference. Original lemma 1.15: sources/sigma-2011-069-askey-wilson/pinned-original-source.tex, source lines 902–902. Navigation pointer for \ref{lem:4} used inside the counted span (line 902); the argument that consumes it is retained in the span.

Lemma 0.1. *The action of* $\mathrm{PSL}_2(\mathbb{Z})$ *on* Λ *is faithful.*

Proof. Pick an integer $n \geq 1$. Consider an element $\eta \in \Lambda$ of the form $\eta = \eta_1\eta_2\cdots\eta_n$ such that for $1 \leq i \leq n$, $\eta_i = s$ for one parity of i and $\eta_i \in \{p, p^{-1}\}$ for the other parity of i. We show that $\eta \notin Z(\Lambda)$. To this end we assume $\eta \in Z(\Lambda)$ and get a contradiction. Let ℓ denote the number of times s occurs among $\{\eta_i\}_{i=1}^n$. Assume for the moment $\ell = 0$. Then $n = 1$ so $\eta = \eta_1 \in \{p, p^{-1}\}$. The elements p, p^{-1} are not in $Z(\Lambda)$, for a contradiction. Therefore $\ell \neq 0$. From the nature of the matrices p, s in Definition 1.14, we may view η as a polynomial in λ that has coefficients in $\mathrm{Mat}_2(\mathbb{F})$ and degree at most ℓ. Call this polynomial f. We claim that the degree of f is exactly ℓ. To prove the claim, write

$$s = s_0 + s_1\lambda, \qquad s_0 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad s_1 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Let $m \in \mathrm{Mat}_2(\mathbb{F})$ denote the coefficient of λ^ℓ in f. The matrix m is obtained from $\eta_1\eta_2\cdots\eta_n$ by replacing each occurrence of s by s_1. Using $s_1ps_1 = -s_1$ and $s_1p^{-1}s_1 = s_1$ we find $m \in \{\pm p^i s_1 p^j \,|\, -1 \leq i, j \leq 1\}$. The matrix p is invertible and $s_1 \neq 0$ so $p^i s_1 p^j \neq 0$ for $-1 \leq i, j \leq 1$. Therefore $m \neq 0$ and the claim is proved. Let κ denote the $(1,1)$-entry of the matrix η. Then $\eta = \kappa I$ since $\eta \in Z(\Lambda)$. In the equation $\eta_1\eta_2\cdots\eta_n = \kappa I$ take the determinant of each side and use Lemma 1.15(i) to get $(-\lambda)^\ell = \kappa^2$. Therefore ℓ is even and $\kappa = \pm\lambda^{\ell/2}$. Now $\eta = \pm\lambda^{\ell/2} I$, so the above polynomial f has degree $\ell/2$. But $\ell \neq 0$ so $\ell > \ell/2$ for a contradiction. the result follows. □

References

[1] Alperin R.C., Notes: $\mathrm{PSL}_2(\mathbb{Z}) = \mathbb{Z}_2 \star \mathbb{Z}_3$, *Amer. Math. Monthly* **100** (1993), 385–386.